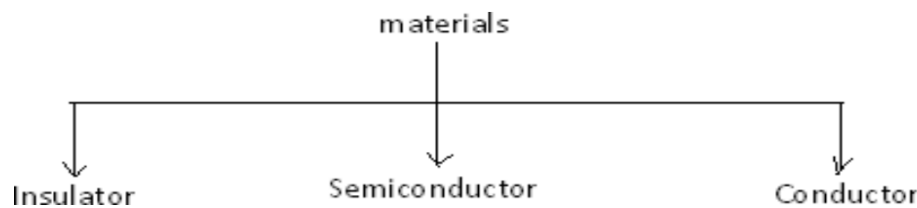


Chapter 2

INTRODUCTON

Based on the electrical conductivity all the materials in nature are classified as **insulators**, **semiconductors**, and **conductors**.



1- INSULATOR :

Is a material that prevents the flow of electricity. Unlike conductors, insulators have extremely high resistance to electrical current because their electrons are tightly bound to their atoms.

1. Characteristics

- **Low Conductivity:** They offer negligible electrical flow when voltage is applied.
- **High Resistivity:** Typical values range between 10^{10} and $10^{12} \Omega \text{ cm}$.
- **Examples:** Glass, paper, mica, and quartz.

2- CONDUCTOR :

Is a material that allows electricity to flow through it very easily. Because conductors have extremely high conductivity, they are used to build wires and electronic components.

Characteristics

- **High Conductivity:** They support a "generous" (large) flow of electrical charge when a voltage is applied.
- **Low Resistivity:** Their resistance to current is very small, typically between 10^{-4} and $10^{-6} \Omega \text{ cm}$.

- **Examples:** Copper, Aluminum, Silver, and Gold.

3- SEMICONDUCTOR :

Is a material that sits between a conductor and an insulator. Its ability to conduct electricity can be "turned on" or "controlled," which is why semiconductors are the foundation of all modern electronics (like chips and processors).

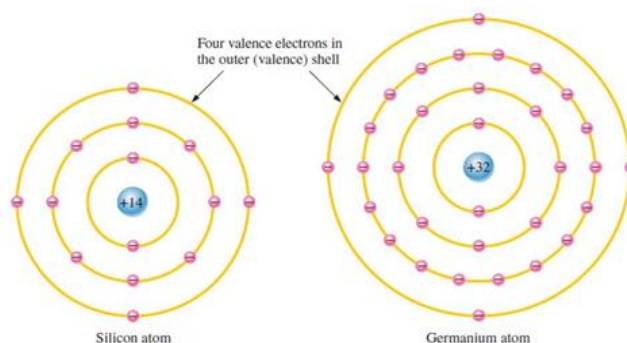
Characteristics

- **Intermediate Conductivity:** Its ability to conduct is better than an insulator but not as high as a conductor.
- **Resistivity:** Typically ranges between 10 and $10^{+4} \Omega \text{ cm}$.
- **Primary Examples:** Two of the most commonly used are Silicon (Si=14 atomic no.) and germanium (Ge=32 atomic no.). Both have 4 valance electrons

II- Review of the simplified description of atomic structure

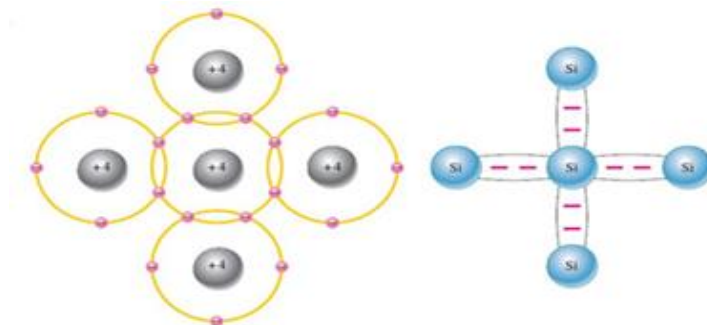
The diagram illustrates the Bohr model for two specific atoms:

- **Silicon Atom Si:**
 - Nucleus: Contains a net positive charge of +14 (representing 14 protons).
 - Electron Shells: It has three orbiting shells. The electrons are distributed as 2 in the first shell, 8 in the second, and 4 in the outer shell.
- **Germanium Atom Ge:**
 - Nucleus: Contains a net positive charge of +32 (representing 32 protons).
 - Electron Shells: It is larger, with four orbiting shells. Despite having more electrons total, it also has exactly 4 electrons in its outer shell.



II-1 Atomic Structure and Covalent Bonding in Intrinsic Semiconductors

The formation of a semiconductor material starts with the **atomic structure** of elements like **Silicon** or **Germanium**, which possess **four valence electrons** in their outer shell. These atoms link together through **covalent bonding**, sharing electrons with four neighbors to create a stable crystal lattice. In this pure state, the material is known as an **intrinsic semiconductor**. An **intrinsic semiconductor** has a very low conductivity at room temperature because almost all electrons are locked in bonds, meaning there are very few carriers that **can easily move**.



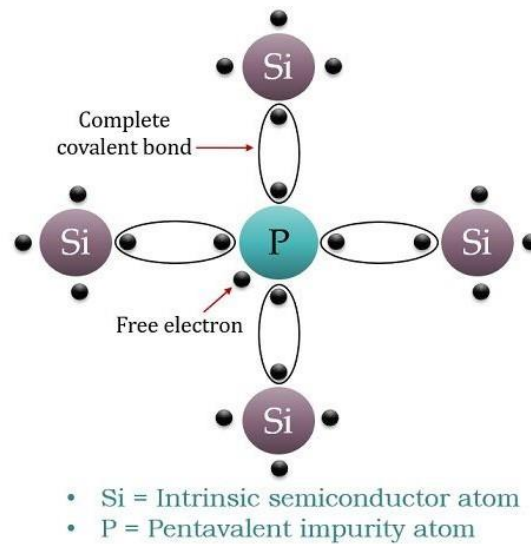
II-2 Doped or extrinsic semiconductors

To make the material useful for electronics, this **intrinsic** base is modified through **doping** to create extrinsic semiconductors. By adding Group V impurities, we create an **N-type** material with extra electrons, while Group III impurities create a **P-type** material with "holes". The final functional structure is formed at the **PN junction**, where these materials meet and create a **depletion region** after contact, allowing for the control of electrical current.

II-2-a Formation of N-Type Semiconductor

N-type extrinsic semiconductors are created when group V elements like Arsenic (As), Phosphorus (P), Antimony (Sb), or Bismuth (Bi) etc., are doped to a complete semiconductor crystal. These are named so because doping these elements will create an additional electron in the atom's valence shell.

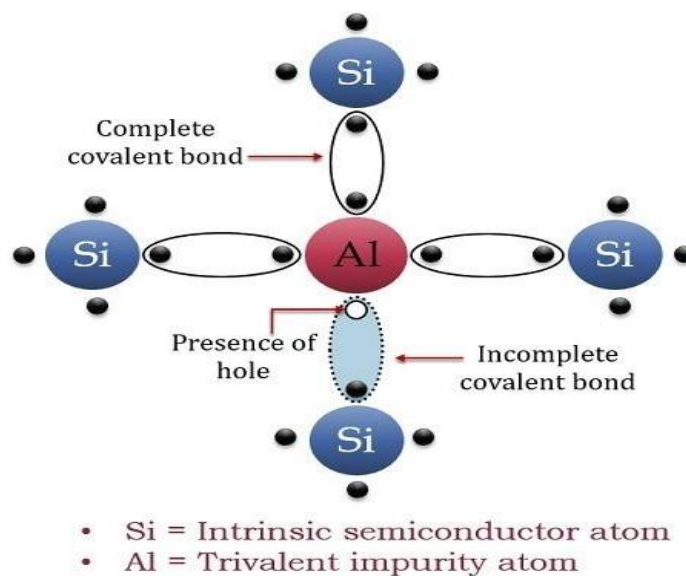
The figure below depicts the doping of phosphorus element into a pure silicon material:



II-2-b P-Type Semiconductor

P-type extrinsic semiconductors are created when **group III elements** like **Boron (B)**, **Aluminum (Al)**, **Gallium (Ga)**, or **Indium (In)** are doped into a pure semiconductor crystal. These are named so because doping these elements will create a **missing electron** (called a **hole**) in the atom's valence shell.

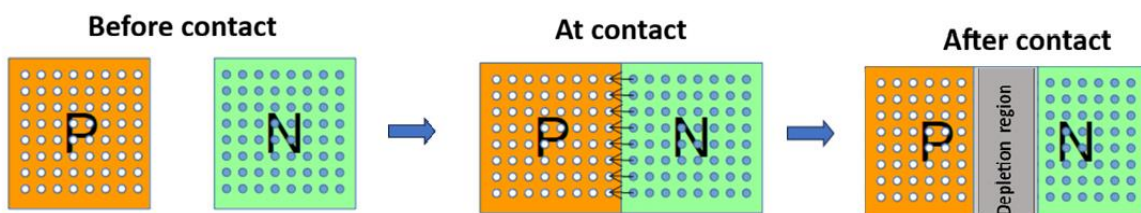
Let us have a look at the below figure that describes doping of aluminum to a pure silicon material:



Parameter	P-Type	N-Type
Impurity doped	Trivalent impurity	Pentavalent impurity
Also known as	Acceptor atom because of presence of additional hole.	Donor atom due to the existence of additional electron.
Doped group	Group III elements. For eg - boron, gallium, indium, aluminium etc.	Group V elements. for eg - arsenic, antimony, bismuth, phosphorus etc.
Majority carriers	Holes	Electrons
Minority carriers	Electrons	Holes
Conductivity	Due to presence of holes.	Due to presence of electrons.

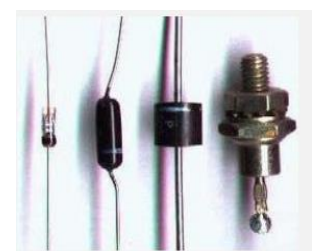
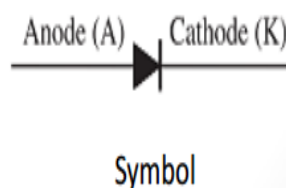
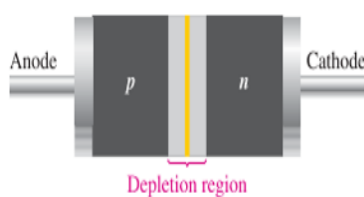
III PN junction

When a P-type semiconductor is brought into contact with an N-type semiconductor, the excess electrons from the N-type rush to fill the holes of the P-type in the contact area. A carrier-free zone is formed across both the P-region (where the holes have been filled) and the N-region (where the electrons have left).



III-1 DIODES

A diode is made from a small piece of semiconductor material, usually silicon, in which half is doped as a P region and half is doped as an N region with a PN junction and depletion region in between.



III-1-1 Biasing

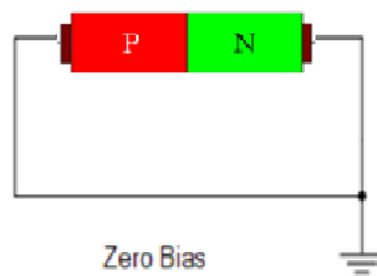
“Biasing” is providing minimum external voltage and current to activate the device to study its characteristics.

There are two operating regions and two "biasing" conditions for the standard Junction Diode, we generally study **three states** to understand the device fully, and they are:

1- Zero Bias ($V=0$):

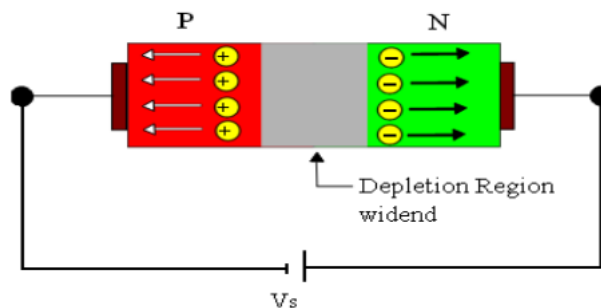
When a diode is Zero Biased no external energy source is applied and a natural Potential Barrier is developed across a depletion layer.

- ✓ Electrons and holes near the junction mix and cancel each other out (recombination).
- ✓ The Result: A depletion region forms in the center, creating an internal electric field (barrier) that stops further movement.
- ✓ Current: There is **no current** flowing through the device.



2. Reverse Bias (Blocked: $V < 0$)

The **Positive** terminal is connected to the **N-type** and the **Negative** to the **P-type**.

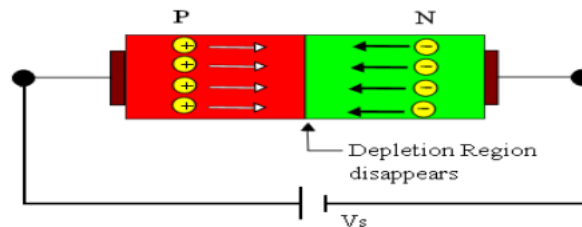


- ✓ The external voltage pulls the charge carriers away from the center, which **widens** the depletion region.
- ✓ The Result: The barrier becomes too large for the majority carriers to cross.

- ✓ Current: The main current is **zero**. Only a tiny "leakage" current exists, called the reverse saturation current (I_s).

3. Forward Bias (Conducting: $V > 0$)

The **Positive** terminal is connected to the **P-type** and the **Negative** to the **N-type**.

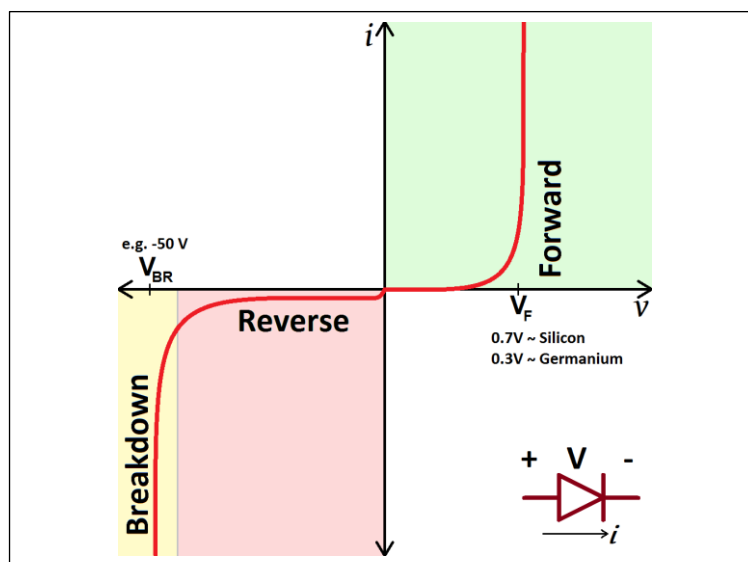


- ✓ The voltage pushes the charges toward the center, which **shrinks** (reduces) the depletion region.
- ✓ The Result: The barrier disappears, allowing electrons to flood across the junction.
- ✓ Current: There is a **very high current** flow that rises exponentially as voltage increases.

III-1-2 Diode I-V Characteristics

Current-Voltage Relationship

The most important diode characteristic is its current-voltage (i - v) relationship. This defines what the current running through a component is, given what voltage is measured across it. Resistors, for example, have a simple, linear i - v relationship...[Ohm's Law](#). The i - v curve of a diode, though, is entirely *non-linear*. It looks something like this:

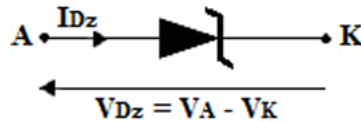


ZENER

IV Zener diode

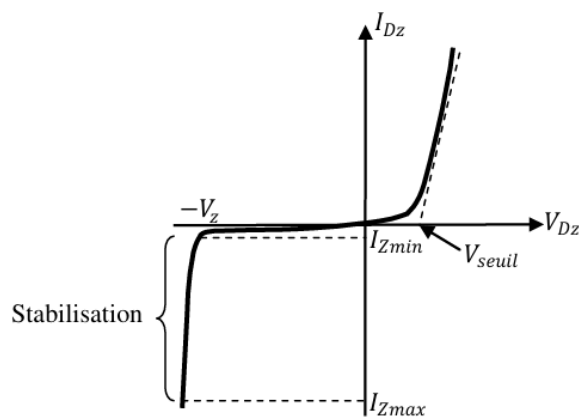
1. Introduction

A Zener diode is a junction diode specially designed to be used under reverse bias in the breakdown region.



Characteristic:

The Zener diode is a non-linear component with the following characteristic:



V_z : Zener voltage of the diode.

$I_z(\min)$: Minimum current below which the voltage is no longer regulated.

$I_z(\max)$: Maximum current that the Zener diode can withstand.

Main application:

For a voltage $V_{Dz} > V_z$, the Zener diode conducts and the voltage across its terminals remains practically constant and equal to V_z .

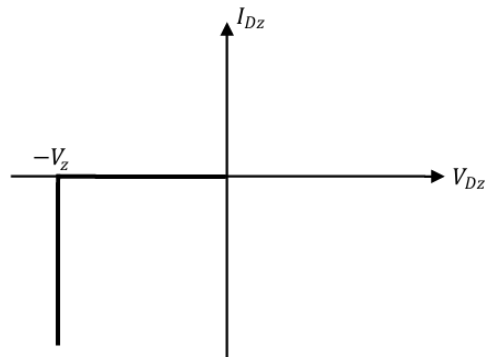
2. Zener Diode Modeling

In forward bias:

The Zener diode behaves like a conventional junction diode.

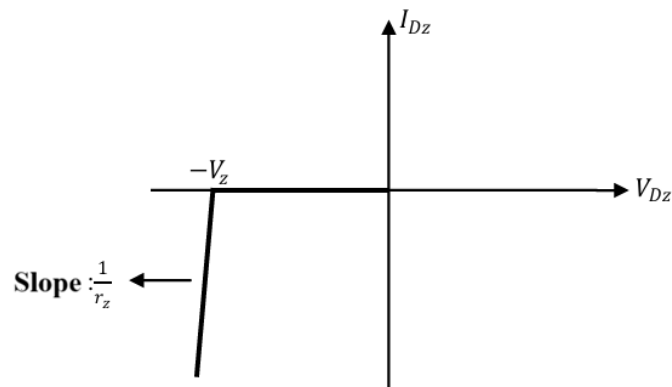
In reverse bias:

- Ideal model



$ V_{Dz} > V_z$	D_z Conducting	
$ V_{Dz} < V_z$	D_z Blocking	

- **Model with Zener voltage and resistance**



$ V_{Dz} > V_z$	D_z conducting	
$ V_{Dz} < V_z$	D_z blocking	

V Rectification, Filtering and Regulation

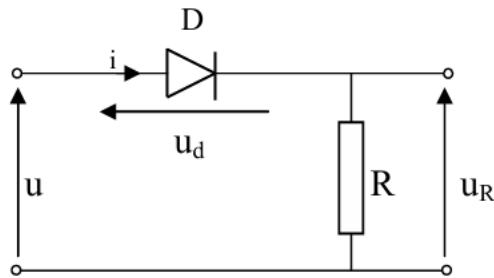
1.Rectification

Definition

Rectification consists of converting a bidirectional voltage into a unidirectional voltage called **rectified voltage**.

-Half-wave Rectification

Circuit Diagram



avec : $u = U_M \sin \omega t$
et $\omega = 2\pi f$

Operating Principle:

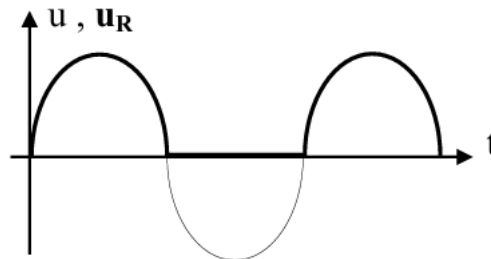
Assumption: The diode is considered ideal.

During the **positive half-cycle** of the voltage u ($u > 0$), the diode **D** is forward biased, therefore it is **conducting** ($i > 0$ and $u_d = 0$), so:

$$u_R = u - u_d = u$$

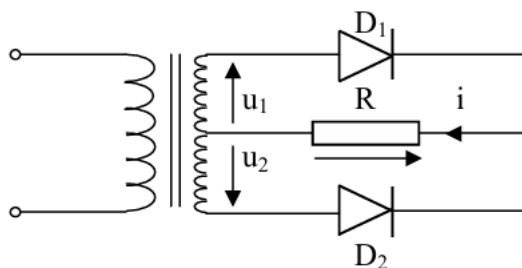
with: $u = U_M \sin \omega t$
and $\omega = 2\pi f$

During the **negative half-cycle** of the voltage u ($u < 0$), the diode **D** is reverse biased, therefore it is **blocked** ($i = 0$ and $u_d < 0$), so: $u_R = 0$



2. Full-Wave Rectification Using Two Diodes and a Center-Tapped Transformer

Circuit Diagram



$$u = u_1 - u_2 = U_M \sin \omega t \quad \text{and} \quad \omega = 2\pi f$$

$$u_1 = \frac{u}{2} \quad \text{et} \quad u_2 = -\frac{u}{2}$$

Operating Principle:

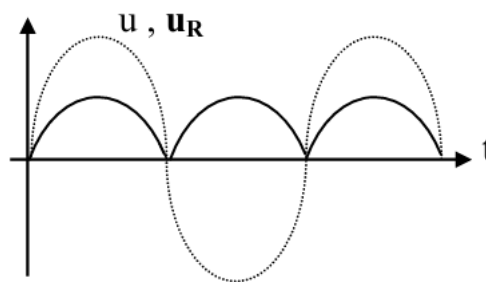
Assumption: The diodes are considered ideal.

During the **positive half-cycle** of u :

- u_1 is positive, D_1 conducts, therefore $u_R = u_1 = u/2$
- u_2 is negative, D_2 is blocked.

During the **negative half-cycle** of u :

- u_2 is positive, D_2 conducts, therefore $u_R = u_2 = -u/2$
- u_1 is negative, D_1 is blocked.

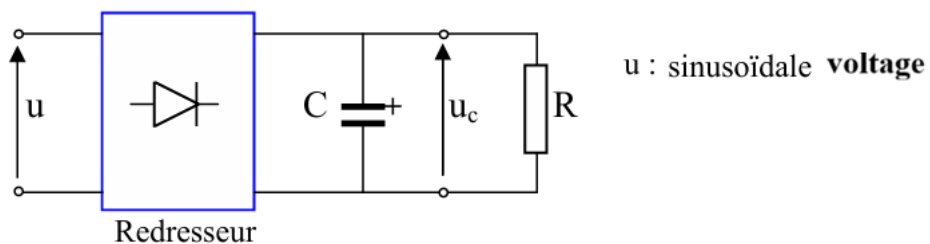


2, Filtering

Definition

Filtering of a rectified voltage consists of reducing the ripple as much as possible in order to obtain a voltage that is as constant as possible. This function can be achieved using a capacitor.

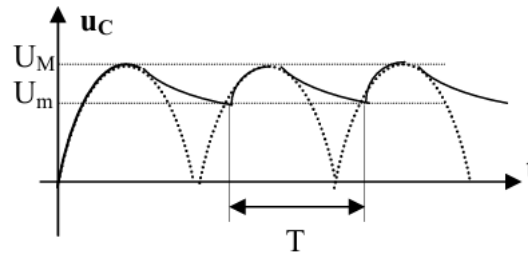
Circuit Diagram



Operating Principle

From the first half-cycle, the capacitor C charges. Then, as soon as the voltage across its terminals becomes greater than the rectified voltage, it discharges through the resistor R .

Waveform of the voltage u_c for a full-wave rectifier



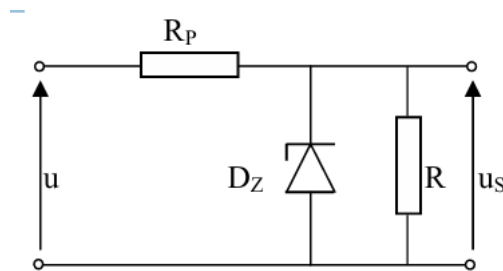
3- Regulation

Definition

Regulation of a ripple voltage consists of obtaining an almost constant voltage. This function can be achieved using a **Zener diode**.

Regulation Using a Zener Diode

Circuit Diagram



u : ripple voltage

R_P : bias resistor of the Zener diode

Operating Principle

- Assume that the load resistor R is disconnected and that the Zener diode resistance R_Z is neglected ($R_Z = 0$).

If $u > U_Z$, then: $u_S = U_Z$

If $u < U_Z$, then: $u_S = u$

Therefore, **u must be greater than U_Z** so that the output voltage remains constant (**regulated**).

- When the resistor R is connected and the Zener resistance R_Z is not neglected,